

LLM-Guided VQC Design

Building a Rigorous, Budget-Matched Benchmark

Interim update — classical baselines complete; the real-LLM matrix is blocked on API quota, not on budget approval

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QUESTION

Does LLM guidance actually help — compared with what?

Primary

At the same unique-candidate evaluation budget, does LLM-guided search beat random, evolutionary and greedy search?

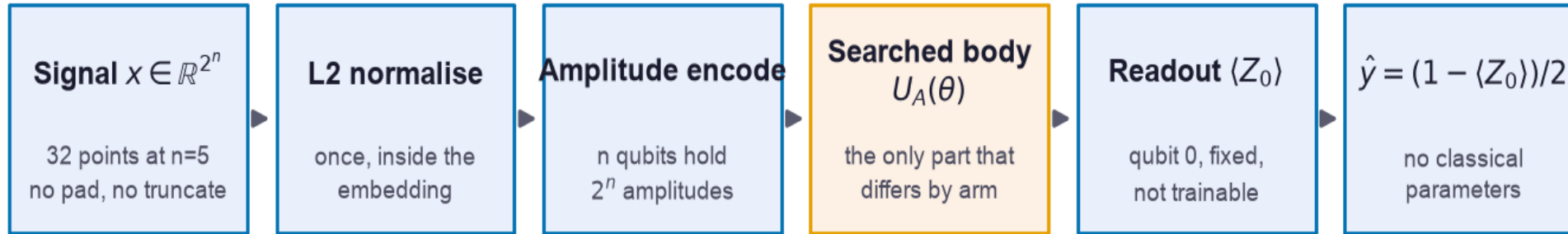
Secondary

How do the selected circuits compare with fixed reference ansätze in predictive error and compiled resource cost?

- Budget matching applies to the search arms. Fixed ansätze are anchors: they are evaluated through the identical training and test pipeline, but they do not run a search.
- Anytime improvement alone proves nothing — random search also improves with more evaluations.
- Gap addressed here: prior agentic-VQC work demonstrates feasibility, but does not isolate the value of the proposal strategy against budget-matched classical search with paired replicates and a protected final test.

WHAT IS MEASURED

One pipeline; only the searched body changes



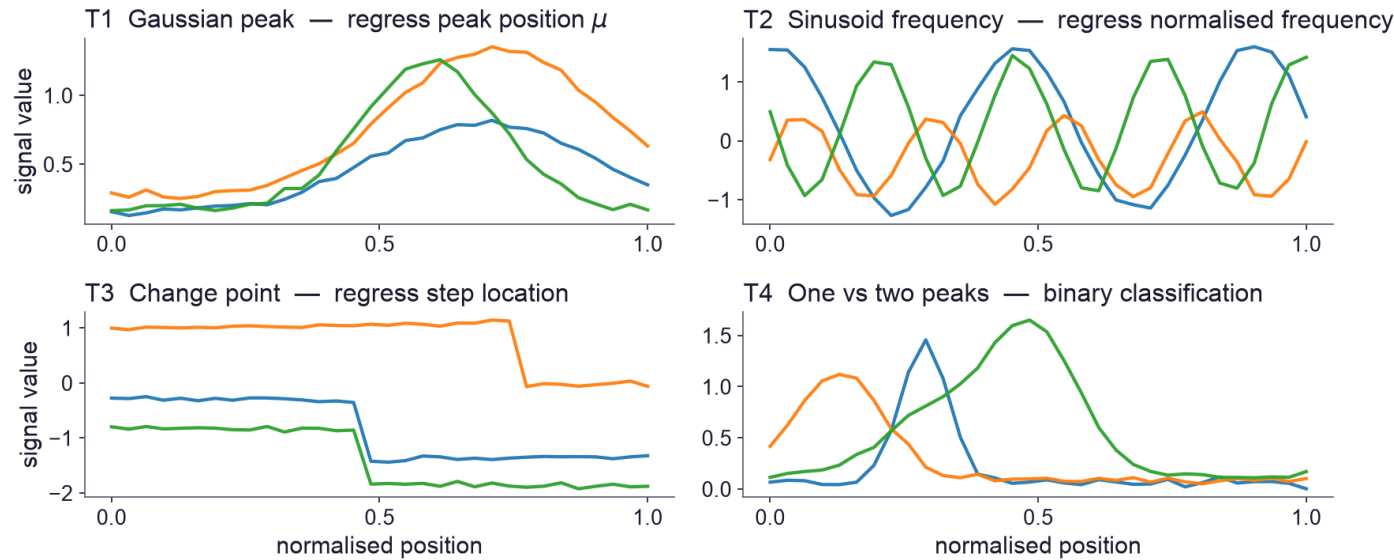
Trained by the shared AdamW loop: only the rotation angles θ of the searched body (40 epochs, batch 32, identical for every arm)

Scored by: validation metric during search → protected-test metric once, after the candidate is selected (the metric is RMSE for T1–T3; T4 is classification)

- A length- 2^n signal fills the amplitudes of n qubits exactly — no padding, no truncation.
- Zero classical parameters: no trainable dense layer anywhere, so a weak circuit cannot be rescued by a classical head.
- Protected-test metric: the same metric formula, re-evaluated once per selected circuit on a final held-out split, after selection is frozen on validation. That formula is RMSE for T1–T3; T4 is classification, so the pipeline-level term is "protected-test metric", never "protected-test RMSE".

TASKS

Four controlled signal tasks, sized for amplitude encoding



2ⁿ points exactly

32 values at $n=5$; the signal is the amplitude vector.

Nuisances randomised

Amplitude, phase, baseline, width and noise vary, so no fixed convention leaks the label.

Scaling

T1 and T2 also run at $n = 3, 4, 6$ and 8.

Splits

256 train / 256 validation / 2048 protected test, per replicate.

Search-space constraint	Value
Qubits n	5 for T1–T4; T1 and T2 also at $n = 3, 4, 6, 8$
Allowed gates	rotations RX / RY / RZ / H · entanglers CNOT / CZ / CRX / CRY / CRZ over line, ring, star, pairs or all-to-all
Max layer operations	$\text{max_ops} = \min(2n, 16)$ — 10 at $n=5$, 16 at $n=8$
Operation $\neq$ physical gate	one operation expands to many physical gates: a rotation layer becomes n single-qubit gates, an entangling layer up to n two-qubit gates

Two tracks separate architecture quality from angle luck

Track A — structure search (primary)

- Arms propose structure only
- One shared AdamW loop trains θ : 40 epochs, batch 32, identical for every arm
- θ -seed derived from the structural hash, so the same circuit gives the same result in any arm

Track B — joint proposal (ablation)

- Candidates carry gates, wires and numeric angles
- θ is evaluated verbatim — no optimizer exists on this path
- Candidate identity includes θ , so new angles make a new candidate

Matched across arms: unique candidate evaluations, training schedule, data splits, seeds, evaluator, protected-test rule. **Not matched:** wall-clock time, training FLOPs, generated tokens — and fixed references run no search at all.

PROGRESS

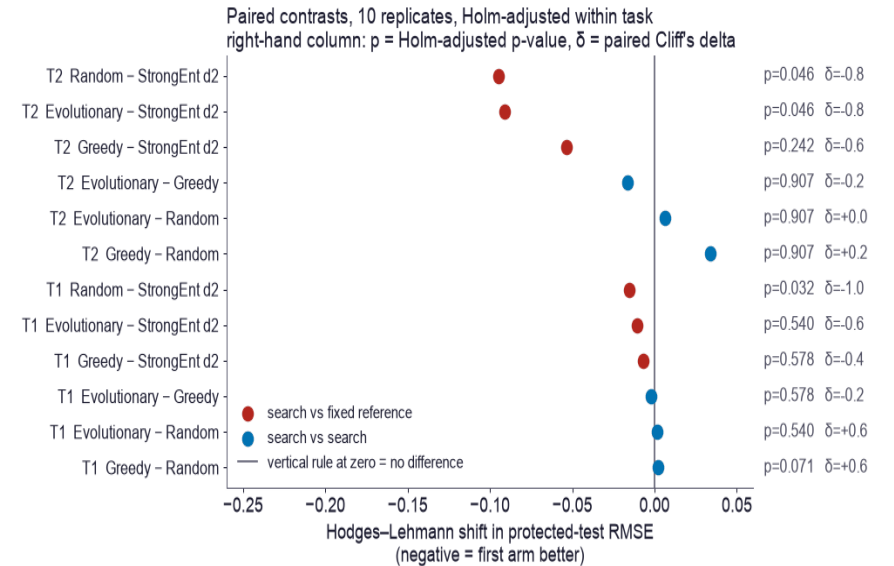
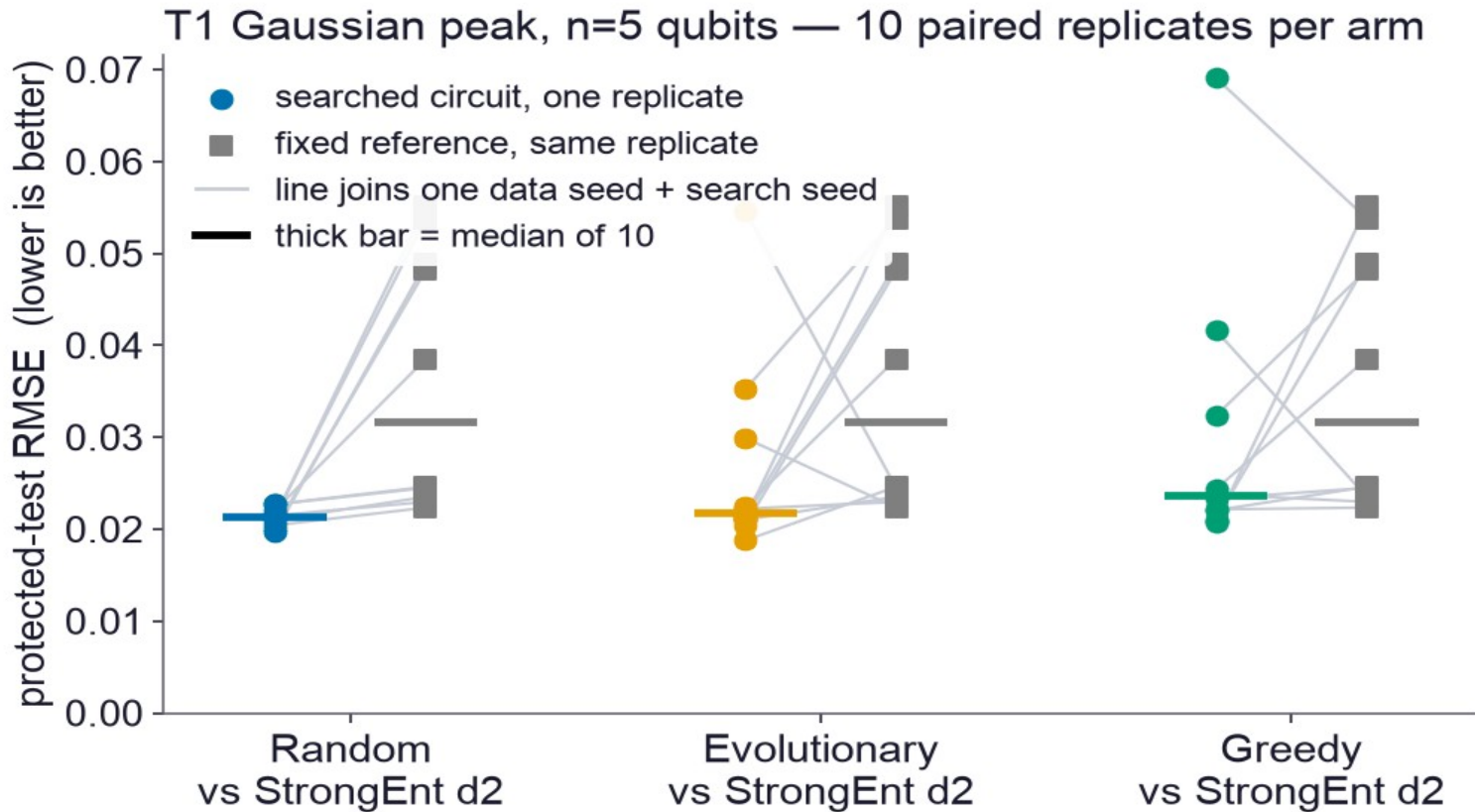
Classical matrix complete; real-LLM matrix not started

Pilot replay		1 / 1 complete
E1 Track B classical		60 / 60 complete
E1 Track B real-LLM		0 / 60 not started — API quota exhausted
E2 Track A classical + refs		280 / 280 complete
E2 Track A real-LLM		0 / 80 not started — API quota exhausted
E3 scaling classical + ref		120 / 120 complete
E3 scaling real-LLM		0 / 80 not started — API quota exhausted
E4 shot / noise robustness		0 / 1 running on the completed classical cells
E5 θ -isolation		0 / 1 blocked — needs LLM-proposed circuits

460 / 460 classical cells complete, 0 failures · 0 / 220 real-LLM cells — the API account returned insufficient_quota, so \$0.00 was spent · **the benchmark as a whole is not complete**

RESULT — 5 QUBITS

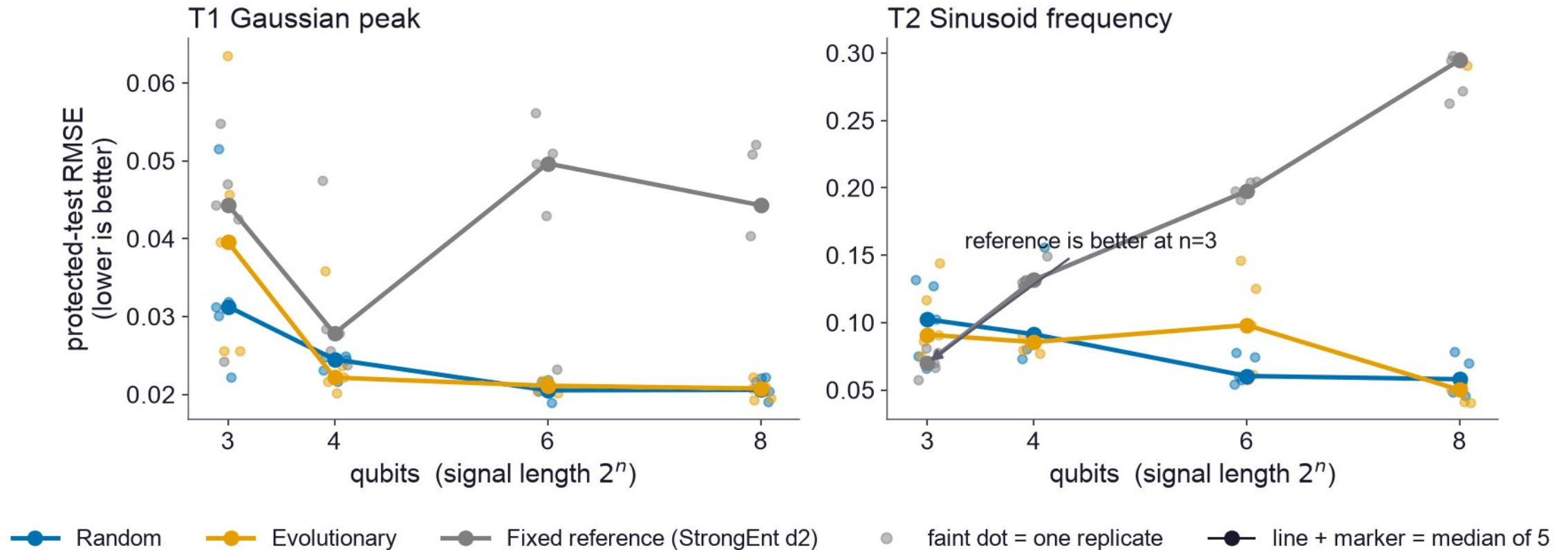
Search beats the fixed reference on T1/T2; no search-method difference was detected



- No Holm-adjusted difference was detected among random, evolutionary and greedy on any task (smallest $p = 0.071$). This is not an equivalence claim.
- Against the strongest reference the picture is task-dependent: on T2 random and evolutionary win ($p = 0.046$, $\delta = -0.8$); on T1 only random is detected ($p = 0.032$) and the shift is small.

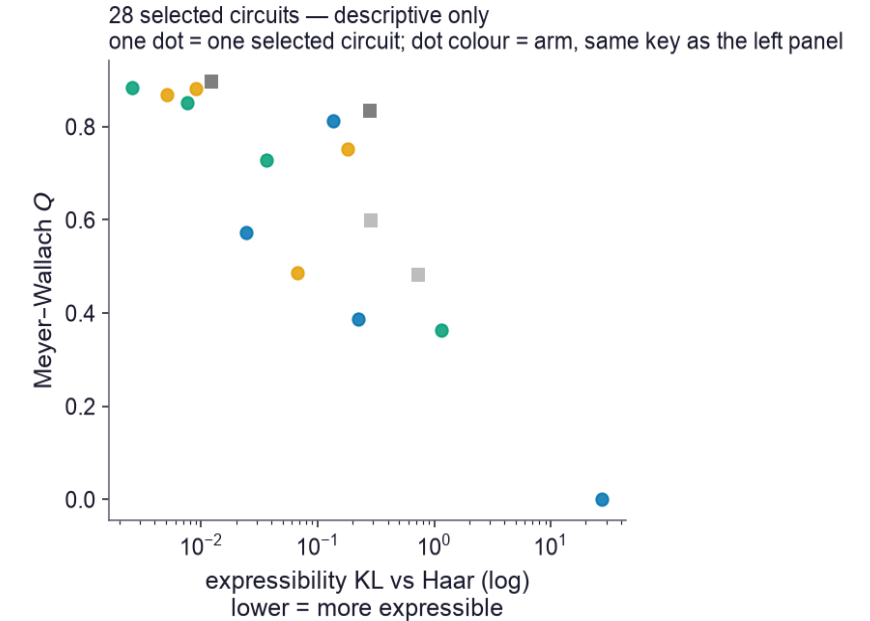
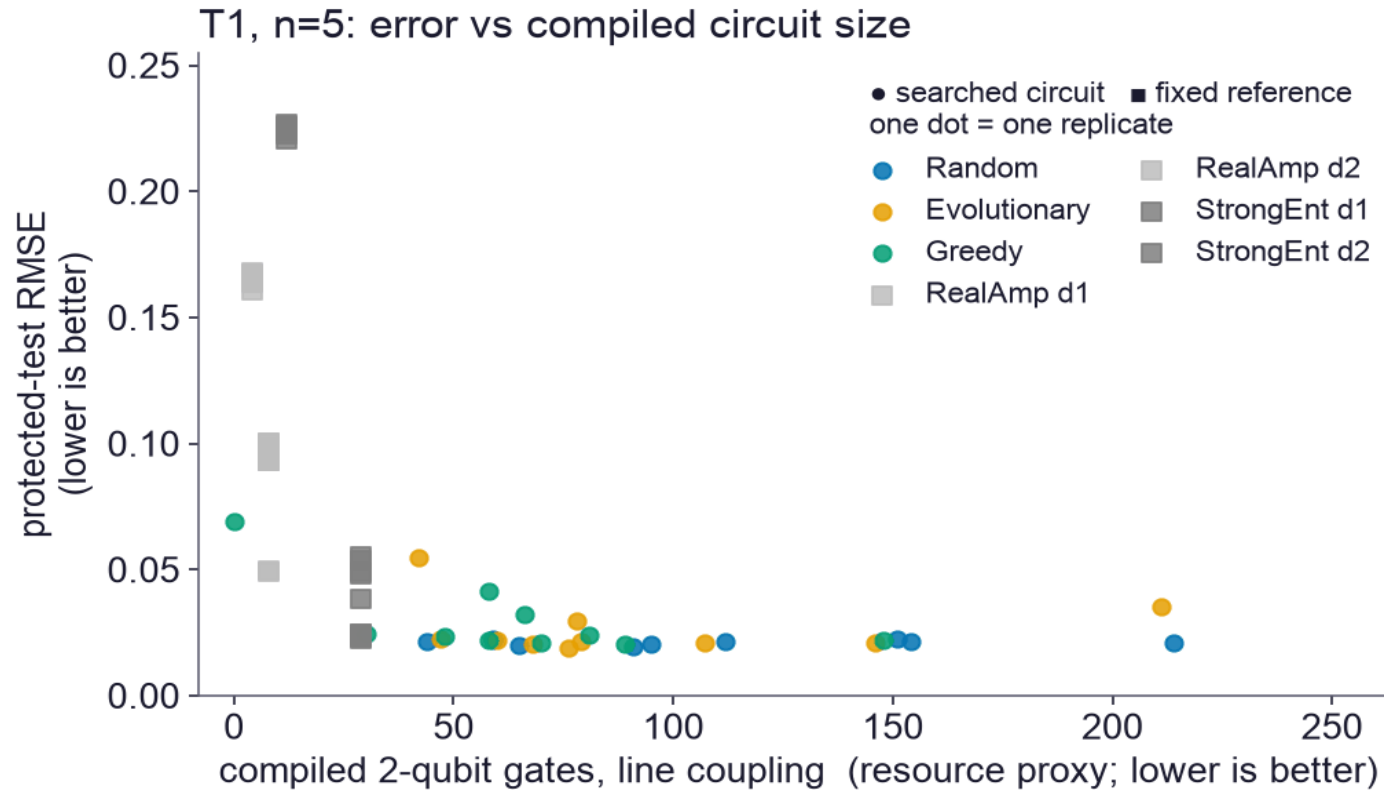
RESULT — QUBIT SCALING

A descriptive crossover: search pulls ahead at larger qubit counts



Descriptive, not confirmatory: at $n=8$ on T2 all 5 paired replicates favour search ($\delta = -1.0$, median 0.058 vs 0.295), but with 5 replicates the smallest attainable p is 0.0625, so after Holm correction across 3 contrasts no result can reach 0.05.

Lower error comes with substantially larger compiled circuits



- Median compiled 2-qubit gates: 58–85 for searched circuits vs 4–29 for the references.
- A hardware-relevant proxy: state preparation is excluded (identical for every arm).
- KL and Meyer–Wallach Q are descriptors; their link to accuracy is untested here.

Interim answer, and what is blocking the rest

Not yet. The LLM hypothesis is still untested. What exists today is a reproducible classical benchmark that defines the bar the LLM has to beat.

Spending control: implemented, not promised

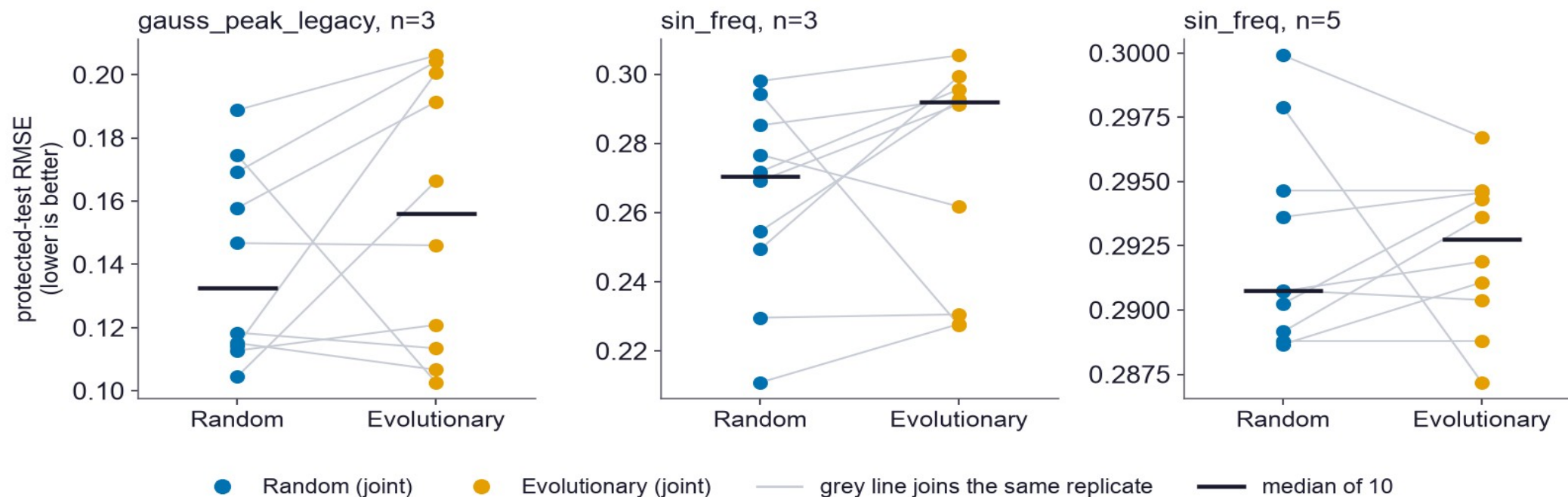
- Durable SQLite ledger shared by every cell, retry and restart — cumulative, not per cell
- Cost settled from returned token counts against a dated price manifest
- Model pinned to gpt-5-nano-2025-08-07; cumulative cap \$2.00
- Measured spend \$0.00 — account returned insufficient_quota, reservations released

Then, in order

- Restore API quota, then run the 220 real-LLM cells (both tracks)
- E5 θ -isolation: is it the architecture or the angles?
- E4 shot-noise and noisy-simulator robustness
- Regenerate the statistical report; completion checker must pass unweakened

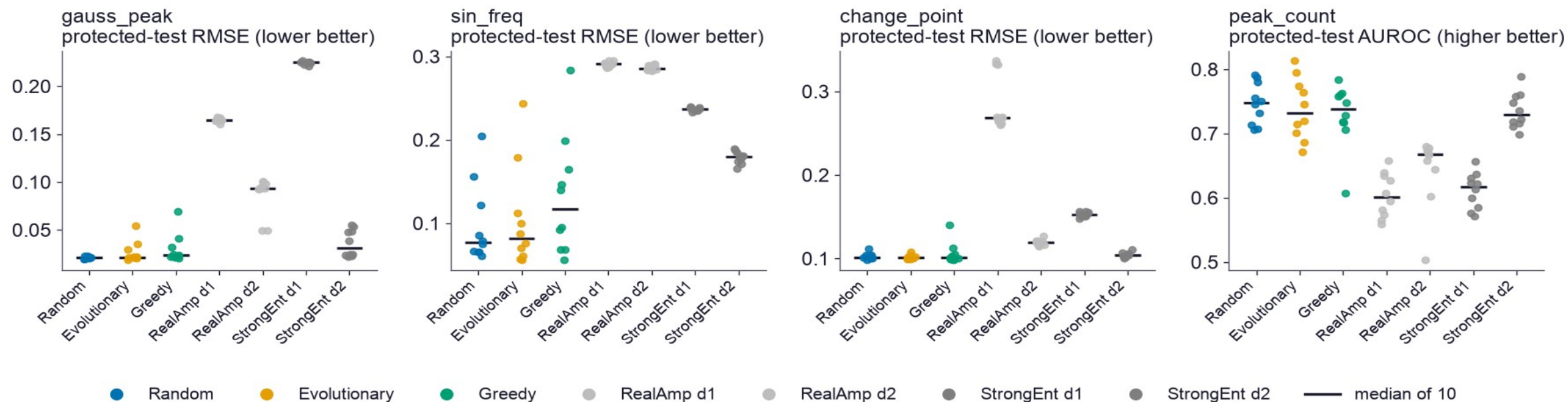
Blocked on API quota, not on approval: cap \$2.00 cumulative, measured spend \$0.00

A1–A2 Pilot replay (E0) and joint search (E1)



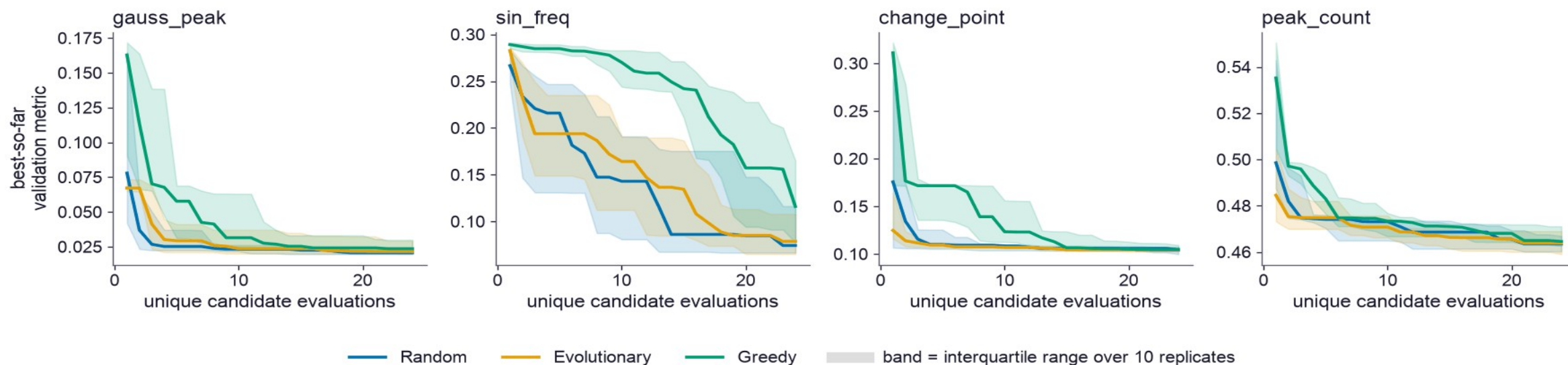
- E0: all 21 candidates of the committed 2-seed real-LLM pilot were re-evaluated offline from stored operations and angles — validation and test values matched exactly, with zero API calls.
- E1 (Track B, verbatim θ , $B=16$): no Holm-adjusted difference between random and evolutionary joint search on any of the three task/qubit cells (all $p \geq 0.28$). T2 at $n=5$ sits near 0.29 RMSE for both.

A3 Full E2 results, all four tasks



- T3 (change point) is saturated: every arm lands at RMSE ≈ 0.10 and all adjusted p-values are 1.00.
- T4 (peak classification) shows no arm separation either: AUROC ≈ 0.73 – 0.75 across all arms.

A4 Anytime search behaviour (E2)



- Median and interquartile band of the best-so-far validation metric against unique candidate evaluations.
- Greedy growth is the slowest starter on T2 but converges to the same region by the 24-evaluation budget.

A6, A8 Statistics, seeds and the protected test

- Paired tests within each (task, n) family: exact Wilcoxon signed-rank plus a paired sign-flip permutation test on the mean difference (10,000 resamples, seed 20260729).
- Multiplicity: Holm step-down applied to the permutation p-values inside each family.
- Effect sizes: Hodges–Lehmann shift (median of Walsh averages, RMSE units) and paired Cliff's δ .
- Verified p-value floor: 5 replicates $\Rightarrow$ smallest two-sided $p = 0.0625 \Rightarrow$ smallest Holm-adjusted value ≈ 0.19 across 3 contrasts. 10 replicates $\Rightarrow 0.00195$.
- Seeds: data seed 1000+r, search seed 2000+r, shared by every arm in a comparison; θ -seed derived from (task, data seed, search seed, structural hash).
- Protected-test metric: the same metric formula used for validation, re-evaluated once per selected circuit on a final held-out split of 2,048 examples, after selection has been frozen. $RMSE = \sqrt{\text{mean}((\hat{y} - y)^2)}$ for the regression tasks T1–T3; AUROC for the T4 classification task, which is why the pipeline-level term is "protected-test metric".
- Search modules cannot import the test gate, and the restriction is enforced by an AST test rather than by convention.

A7, A10 Search space, candidate identity, diagnostic selection

- Grammar scalable_layered_v1: ordered list of at most $\min(2n, 16)$ operations — rotation layers (RX/RY/RZ/H) and entangling layers (CNOT/CZ/CRX/CRY/CRZ) over line, ring, star, pairs or all-to-all patterns.
- Encoding and readout are fixed by the profile, never proposed. Every structure must contain at least one trainable rotation.
- Track A identity = structural hash of the canonical IR; Track B identity additionally includes θ , so re-proposing the same shape with new angles is a new candidate.
- Diagnostic circuits: for each of the 4 tasks $\times$ 7 arms in E2, the replicate whose selected candidate had the best validation metric — 28 circuits, selected on validation only.
- Expressibility: KL between the sampled fidelity distribution and the Haar distribution, 200 sampled states and 2,000 fidelity pairs, $\theta \sim U[-\pi, \pi]$, seeded by architecture hash.
- Enforced parity for the LLM arms: one gate per rotation operation, entangler wires "all", and $1..\text{floor}(n/2)$ disjoint non-duplicate pairs — so no arm can pack several gate layers into one operation. A scan of all 25,973 operations produced by the completed classical search arms found zero violations, so the comparison already run is unaffected.

A9 Resource and transpilation assumptions

- Compiler: Qiskit; basis gates [rz, sx, x, cx]; optimization_level 1; seed_transpiler 7; coupling maps line, ring and all-to-all.
- Routing SWAPs are decomposed into cx, so the reported two-qubit count already includes routing overhead.
- Amplitude state preparation is excluded: its decomposition is identical for every arm at a given n and would dominate the body signal. The number is therefore a relative proxy, not a total device cost.
- Logical counts (1-qubit, 2-qubit, controlled-rotation, parameters, depth) come from the shared IR and are recorded for every evaluated candidate.
- Shot-noise (1,024 / 4,096) and the frozen depol_ro_v1 noisy profile belong to E4, which re-evaluates the best-validation selected circuit of every completed E2/E3 cell — it needs no LLM cells, so it runs on the classical matrix.

A11 API cost model and the cumulative cap

Priced from a dated manifest and settled from returned tokens — no flat per-call charge

Quantity	Value	Source
Pinned model	gpt-5-nano-2025-08-07	OPENAI_MODEL, verified before any client is built
Price (dated manifest)	\$0.05 / 1M input, \$0.40 / 1M output	configs/bench_v2/model_prices.json
Cells to run	220 (E1 60, E2 80, E3 80)	protocol_v2.yaml
Candidates per call	3	llm_arms.py BATCH_SIZE
Expected total calls	≈1,480	ceil(B/3) + duplicate allowance, over 220 cells
Expected cost per call	≈\$0.0004	measured prompt size × manifest price
Expected total cost	≈\$0.6	1,480 × \$0.0004
Per-cell call guard	40 calls (runaway guard — NOT the global cap)	llm_providers.py
Cumulative global cap	\$2.00, durable across cells, retries and restarts	GlobalSpendLedger (SQLite, BEGIN IMMEDIATE)
Actual measured spend	\$0.00 — 0 settled calls; account returned insufficient_quota	runs/bench_v2/preflight_ledger.sqlite

Cost is reserved before each request and settled from returned token counts; a claim that would exceed the cumulative cap is refused before the request is issued

A12, A13 Integrity checks and references

- Break-tests: three safeguards were deliberately broken to confirm the tests catch them. Two exposed real blind spots — a plain `import torch.optim` evaded the AST check, and a self-consistent budget break evaded the ledger test — both are now closed.
- Completion contract: 17 machine-checked criteria; currently 10 pass. The failures are correct (missing LLM cells, E4/E5 markers, final report) and the checker was not modified.
- The self-consistent budget break that once evaded the ledger test is now closed twice over: the cumulative SQLite ledger is proved to survive a restart in a separate OS process, and a cap breach is refused before the request is issued rather than detected after it.
- Protocol content is hash-pinned in `outputs/bench_v2/protocol_hashes.json`. There was no external preregistration; the deck therefore says protocol-frozen throughout.
- Key references: Sim et al. 2019 (expressibility, entangling capability); Knipfer et al. 2026 and Sakka et al. 2026 (LLM-driven VQC design); DQAS; QuantumNAS; EVQE and MoG-VQE; TD-QAS; PWMCTS; BenchRL-QAS.